

Collective Field Representation of Nonrelativistic Fermions in (1+1) Dimensions

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Abstract

A collective field formalism for nonrelativistic fermions in (1+1) dimensions is presented. Applications to the D=1 hermitian matrix model and the system of one-dimensional fermions in the presence of a weak electromagnetic field are discussed.

1 Source Metadata

This sample is based on arXiv record [hep-th/9109013](#). Authors: Dimitra Karabali.

2 Extracted Prose 1

A collective field formalism for nonrelativistic fermions in (1+1) dimensions is presented. The quantum mechanical fermionic problem is bosonized and converted to a second quantized Schrödinger field theory. A formulation in terms of current and density variables gives rise to the collective field representation. Applications to the math expression hermitian matrix model and the system of one-dimensional fermions in the presence of a weak electromagnetic field are discussed. =0

3 Extracted Prose 2

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